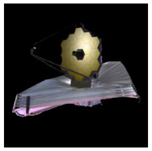

7238 - Constructing High-Dynamic Range PSFs for the MIRI detector for accurately calibrating extended source photometry

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

While the photometric calibration of point sources with MIRI are accurate to within 1% (Gordon et al., 2024), the absolute calibration of *extended sources* - such as deep field extragalactic objects - may be off by up to 7%. This is all due to our lack of understanding of the full spatial extent of the MIRI PSF and the characteristics of its encircled energy curve at large radial distances.

An unfortunate attribute of the Si:As detectors used in MIRI is their low absorption efficiency at shorter (<12 micron) wavelengths, which results in scattering of the photons within the detector. The now well known feature, the cross (or cruciform) artifact (Gaspar et al., 2021), is the easily recognized (but very difficult to model) diffraction pattern produced by the scattering. It is accompanied by a much more evenly dispersed but just as substatial halo. These patterns displace 21% of the stellar flux from the PSF core at 5.6 microns, based on theoretical models. Low SNR and small field of view PSFs do not include these extended features and calibrations determined on - and for - point sources are therefore not accurately

applicable for extended objects.

Here we propose to use archival data to produce high dynamic range PSFs for the MIRI imager at all wavelengths, which will enable accurate photometry of extended objects. The PSFs we will construct will also be used to refine WebbPSF models, thereby also helping to calibrate the LRS and MRS photometry.

OBSERVING DESCRIPTION